Abstract: This paper presents a novel approach to neural machine translation using attention-based transformer architectures. We demonstrate significant improvements in BLEU scores on WMT14 English-German benchmarks compared to previous methods.

1. Introduction

Neural machine translation has revolutionized language processing tasks. Our method builds upon the foundational work of Vaswani et al. (2017) and extends it with multi-head cross-attention layers.

2. Methodology

We train our model on 4.5M sentence pairs using the Adam optimizer with a learning rate of 0.0001 and batch size of 32. The model architecture consists of 6 encoder and 6 decoder layers with 8 attention heads.

3. Results

Our model achieves a BLEU score of 28.4 on WMT14 En-De, surpassing the baseline by 2.1 points. Ablation studies confirm the importance of each component.

4. Conclusion

We have demonstrated a new state-of-the-art result for neural machine translation. Future work will explore larger datasets and multilingual transfer learning.

References

[1] Vaswani, A. et al. (2017). Attention is all you need. NeurIPS 2017.
[2] Bahdanau, D. et al. (2015). Neural machine translation by jointly learning to align and translate. ICLR 2015.